Retail Sales Performance Analysis (DA-001): Project Report

BTech Intro to Data Analytics · Semester III, Section B · Group BT_DA_13 · Mentor: Nitish Sheoran

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Abstract

Total revenue is 4,56,000 from 1,000 orders, an average order value of 456, and three findings stand out: Electronics earns the most revenue (34.4%), May 2023 was the best month, and items priced 300 or more give 88.4% of revenue from 40.3% of units. This project analyses 1,000 retail transactions from January 2023 to January 2024 to find which categories, products and months drive revenue. The data was cleaned and summarised in Python (Pandas) on Google Colab, and the results are shown as KPIs, tables, charts and an HTML dashboard.

1. Introduction

A retail business has sales records but no clear view of which products, categories and periods contribute most to revenue. Store managers need that view to plan inventory, promotions and product assortment. This project turns the raw transaction records into a cleaned dataset, a KPI summary, charts and a short set of evidence-based findings.

2. Objectives

The project has four objectives:

- Calculate total revenue and order volume.
- Compare product and category performance.
- Identify monthly trends.
- Summarise the findings with charts and documented insights.

3. Dataset description

The dataset was downloaded from Kaggle. It holds 1,000 retail transactions in 9 columns, dated 1 January 2023 to 1 January 2024, across three categories: Beauty, Clothing and Electronics.

Column	What it holds
Transaction ID	Unique order number
Date	Date of the sale
Customer ID	Customer identifier
Gender	Male or Female
Age	Customer age (18 to 64)
Product Category	Beauty, Clothing or Electronics
Quantity	Units per order (1 to 4)
Price per Unit	25, 30, 50, 300 or 500
Total Amount	Quantity × price per unit

The dataset does not state a currency, so amounts are shown as plain numbers.

4. Methodology

The analysis follows six steps in a Google Colab notebook, with Pandas for the calculations.

1. Load the CSV and check its shape: 1,000 rows and 9 columns.
2. Inspect data types, missing values and duplicates: there are 0 missing values and 0 duplicate rows.
3. Convert Date to a date type, add a month column, and take revenue from Total Amount (checked equal to Quantity × Price per Unit).
4. Group the data by month, category and gender, and compute orders, units and revenue for each.
5. Because the data has no product names, define a product group as Category @ unit price. This gives 15 groups.
6. Draw charts with Matplotlib and Seaborn, and show the results in an HTML dashboard built with Tailwind CSS and Chart.js.

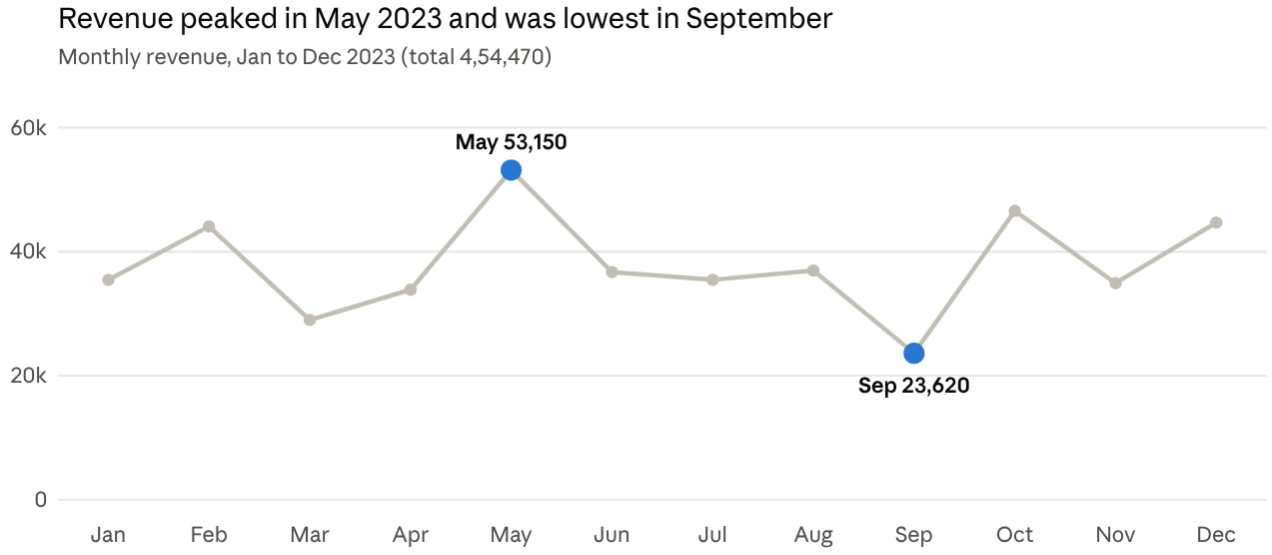
January 2024 has sales on only one day (1,530 from 2 orders). It counts in the totals but is left out of the month comparisons.

5. Results: KPI summary

The store earned 4,56,000 from 1,000 orders, and each order averaged 456 across 2.51 units.

KPI	Value
Total revenue	4,56,000
Total orders	1,000
Units sold	2,514
Average order value	456
Average units per order	2.51
Average price per unit sold	181.38

6. Results: monthly revenue trend



retail_sales_dataset.csv · 1,000 transactions · Jan–Dec 2023 (Jan 2024 left out: 1 day of data)

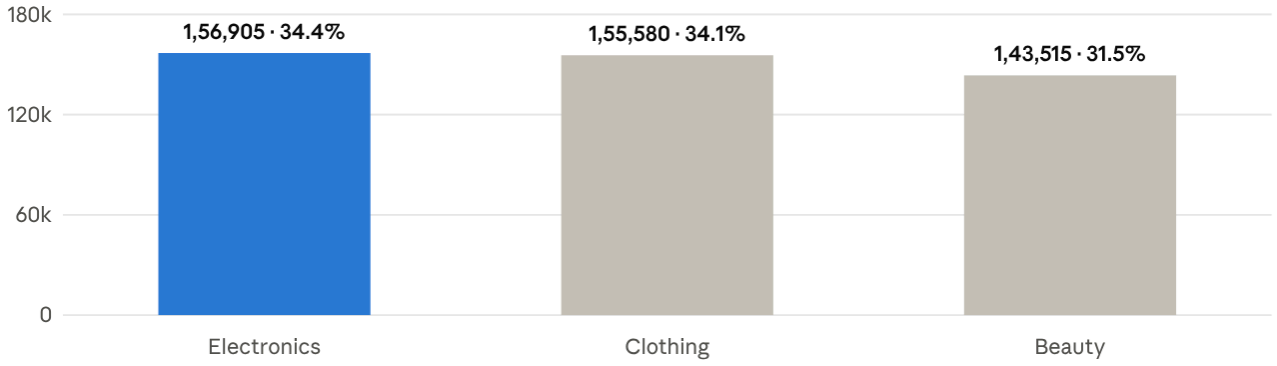
Revenue peaked in May 2023 at 53,150 and was lowest in September 2023 at 23,620; October (46,580) was second best. January 2024 is not shown because it has sales on one day only (1,530).

7. Results: category performance

Electronics earns the most revenue (1,56,905, 34.4%), but the three categories are within 8.5% of each other.

Electronics leads on revenue, but the three categories are close

Revenue by category, Jan 2023 to Jan 2024 (total 4,56,000)



retail_sales_dataset.csv · 1,000 transactions · Jan 2023 to Jan 2024

Category	Revenue	Share of revenue	Units	Orders
Electronics	1,56,905	34.4%	849	342
Clothing	1,55,580	34.1%	894	351
Beauty	1,43,515	31.5%	771	307

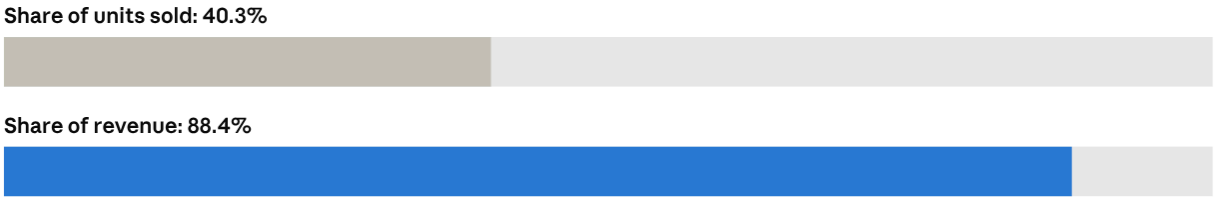
Clothing has the most orders (351). By gender, female customers account for 2,32,840 (51.1%) of revenue and male customers for 2,23,160 (48.9%).

8. Results: product groups and price tiers

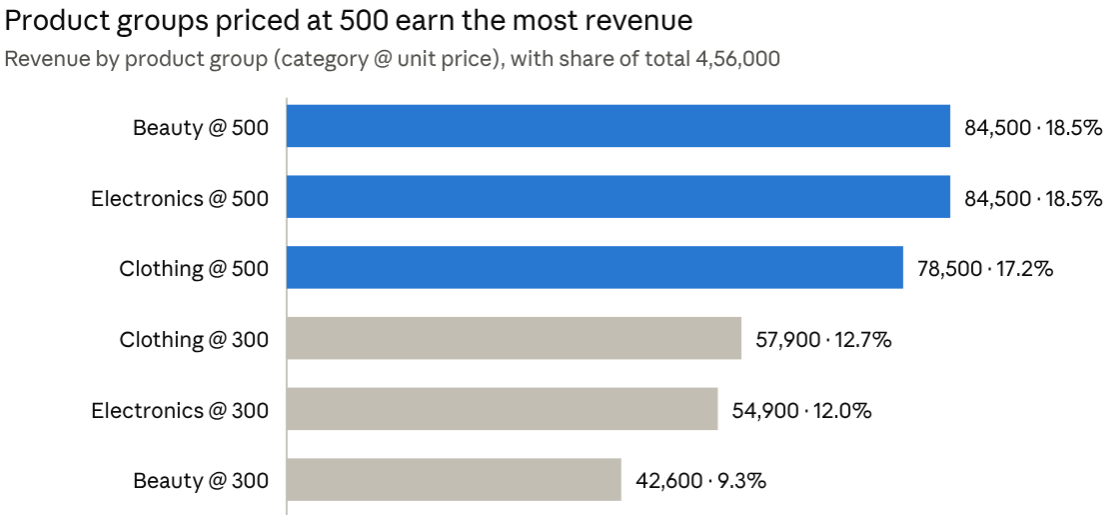
Items priced 300 or more are 40.3% of units sold but 88.4% of revenue, and they are exactly the six product groups drawn below. The top 10 of the 15 product groups together earn 95.3% of revenue.

Items priced 300 or more are a minority of units but most of revenue

Six product groups with unit price 300 or 500, as a share of all sales



retail_sales_dataset.csv · 1,000 transactions · units 1,013 of 2,514; revenue 4,02,900 of 4,56,000



retail_sales_dataset.csv · 1,000 transactions · top 6 of 15 product groups

These six groups sold 1,013 of the 2,514 units (40.3%).

A product group is Category @ unit price, because the dataset has no product names.

9. Key insights

1. Electronics is the top category (34.4% of revenue) and Beauty the lowest (31.5%), so no category dominates.
2. Clothing has the most orders (351), followed by Electronics (342) and Beauty (307).
3. May 2023 was the best month (53,150) and September 2023 the weakest (23,620).
4. Items priced 300 or more are 40.3% of units sold but 88.4% of revenue.
5. Average order value is 456; Beauty is the highest (467) and Clothing the lowest (443). Revenue is split fairly evenly between female (51.1%) and male (48.9%) customers.

10. Recommendations

These suggestions follow from the findings above and are for store managers to weigh.

- Keep items priced 300 and 500 well stocked, as they give 88.4% of revenue.
- Plan promotions for weaker months such as September.
- Bundle low-price items (25 to 50) with higher-priced ones to lift order value.
- Keep the assortment balanced across the three categories, as none dominates.

11. Limitations and future scope

This is a descriptive analysis of one year of data, so it shows patterns rather than causes.

- Only one year of data is available, so no claim about seasonality can be made.
- Product groups are built from category and unit price, not from real product names.
- January 2024 has one day of sales and is left out of the month comparisons.
- The dataset does not state a currency.

Future work could add customer-level analysis (repeat purchases, age groups) and forecasting once more months of data are available.

12. Conclusion

Revenue in this store is spread almost evenly across Electronics, Clothing and Beauty, but it is concentrated in higher-priced items: those priced 300 or more give 88.4% of revenue from 40.3% of units. May 2023 was the strongest month and September 2023 the weakest. These findings answer the project's objectives and give store managers a clear basis for planning stock and promotions.

13. Tools and references

- Python 3, Pandas, NumPy, Matplotlib and Seaborn, run in Google Colab.
- HTML dashboard built with Tailwind CSS and Chart.js.
- Dataset: retail sales dataset downloaded from Kaggle.
- Project code and dashboard: GitHub repository.
- Project brief: DA-001, Retail Sales Performance Analysis, BTech Intro to Data Analytics (mentor: Nitish Sheoran).